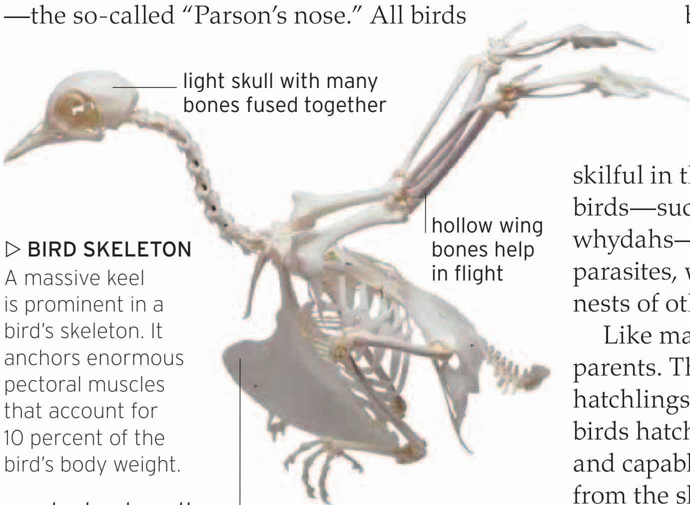
BIRDS

Birds are perhaps the most strikingly conspicuous of all land vertebrates. The first feathered birds evolved from a small group of hollow-boned dinosaurs. These warm-blooded flying animals went on to become one of the most species-rich of vertebrate classes—many with dazzling colors or melodious calls. Today, most birds have a high-speed metabolism suited for a frenetic lifestyle. Some rank among the fastest vertebrates on the planet and they combine speed with impressive brain power to find food and raise a family.

Anatomy

Birds are unique among vertebrates in many respects. Their skin is feathered, their bones contain air spaces to make them lightweight, and their tail vertebrae are fused into a stump —the so-called "Parson's nose." All birds walk, run, or perch on their hindfeet, while their forelimbs are adapted as wings—with the wrist and "hand" modified for greater rigidity. A few birds are flightless, but the vast majority are aerobatic and have a prominent keel on their breastbone for supporting the massive muscles needed to flap. Tiny hummingbirds, beating their wings up to 50 times per second, can even fly backward. A bird's head contains a large brain and big eyes—and is supported by a long, flexible neck with more neck vertebrae than in mammals. Its jaws are toothless and have a horny covering that makes up the bill. The shape of the bill varies a great deal, depending on dietary habits—for example, sharply hooked in predators, strong and stubby in seedeaters.



▷ BIRD SKELETON
A massive keel is prominent in a bird's skeleton. It anchors enormous pectoral muscles that account for 10 percent of the bird's body weight.

Like mammals, birds have a warm-blooded body and a strong, four-chambered heart. In order to maximize oxygen intake, their respiratory system also has a complex arrangement of sacs in the chest and abdomen, which helps flush stale air out of the lungs and replace it with fresh air.

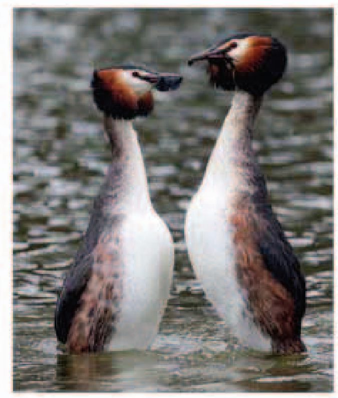
Reproduction

Birds are the only class of vertebrates that are exclusively egg-laying. Their eggs have hard, chalky shells and extra yolk to support the developing embryo. Virtually all birds take advantage of their warm-bloodedness to incubate their eggs, and many build elaborate nests to house them. Some, such as weaver birds, are particularly skilful in their nest-building. However, a few birds—such as cuckoos and the finchlike whydahs—have evolved to be brood parasites, which lay their eggs in the nests of other species.

Like mammals, most birds are dedicated parents. They care for both their eggs and hatchlings. More primitive ground-dwelling birds hatch precocious chicks that are feathered and capable of running soon after breaking from the shell. But most birds, including almost all tree-dwellers, hatch "altricial" chicks, which are born naked, blind, and entirely dependent on the parents to provide them with food.

▽ CATCHING PREY
Flight helps birds escape danger, while turning some into stealthy predators. The flight feathers of owls are fringed to muffle any sound as they swoop down on prey.

▷ DANCE DISPLAY
Bird courtship involves displays of song or even–as in great crested grebes–displays of dance. Such rituals help seal the cooperative bonds needed to raise a family.



Behavior

Birds exhibit complex behavior that is made possible by good senses of sight and hearing. The higher parts of their brain—including the cerebellum, the part involved in the coordination of complex movements and important for flying—are especially well developed. Many combine flight with impressive navigational skills to accomplish long-distance migrations. The Arctic tern's migration—longer than that of any other bird —takes it between the Arctic and Antarctic summers every year. It sees more daylight than any other animal. Some birds demonstrate skills that can only be developed by learning and a few—like some mammals—even use tools to manipulate their environment and find food.

Birds vary widely in the type and extent of their social behavior. Many use elaborate courtship displays to find a mate—showing off with colorful plumage, rich calls and songs, or even ritual dances. Breeding for most birds is a private, monogamous affair—even for species that flock together for the rest of the year. But others—such as many seabirds —nest communally, raising their families in noisy, crowded rookeries.